

Problem 1 — Sum of squares

$S(n) = 1^2 + 2^2 + \cdots + n^2$

Step 1

n	0	1	2	3	4
$S(n)$					

Step 2 How is $S(n)$ related to $S(n - 1)$?

Step 3 Base case:

Recursive case:

Step 4 Check: $S(10) = 385$.

call	expands to	value
$S(4)$	$S(3) + 16$	30
$S(3)$	$S(2) + 9$	14
$S(2)$		
$S(1)$		
$S(0)$	base case	0

Problem 2 — Sum of the first n odd numbers

$O(n) = 1 + 3 + \cdots + (2n - 1)$

Step 1

n	0	1	2	3	4
$O(n)$					

Step 2 How is $O(n)$ related to $O(n - 1)$?

Step 3 Base case:

Recursive case:

Step 4 Check: $O(20) = 400$.

call	expands to	value
$O(4)$	$O(3) + 7$	16
$O(3)$	$O(2) + 5$	9
$O(2)$		
$O(1)$		
$O(0)$	base case	0

Problem 3 — Geometric series

$G(r, n) = 1 + r + r^2 + \cdots + r^n$

Step 1

n	0	1	2	3	4
$G(2, n)$					

Step 2 How is $G(r, n)$ related to $G(r, n - 1)$?

Step 3 Base case:

Recursive case:

Step 4 Check: $G(2, 10) = 2047, G(3, 4) = 121$.

call	expands to	value
$G(2, 3)$	$1 + 2 \cdot G(2, 2)$	15
$G(2, 2)$	$1 + 2 \cdot G(2, 1)$	7
$G(2, 1)$		
$G(2, 0)$	base case	1